BookFlix

Web App Development project documentation

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1. Introduction

BookFlix is a web application that works as an online bookstore and a reading platform. Visitors can browse a catalog of 49 books, search it and filter it, and after creating an account they can buy books, read them online, keep track of their reading and write reviews.

Besides the usual shop features, the application has a few features built on Google Gemini that are meant to be used while reading a book: an interactive map of the book's world, conversations with its characters and a detective game for crime novels. All of them take into account how far the reader has read, so they do not reveal anything from later chapters.

The application also has an administrator account, which can manage the catalog and moderate reviews from an admin page.

2. Technologies Used

The project is split into three parts: the frontend written in React, Firebase for user accounts and data, and a small Node.js server that handles the AI requests.

Technology	**What it is used for**
React 19	The user interface. The pages are built from components, and state is handled with hooks such as useState, useEffect and useContext.
React Router 7	Navigation between pages, routes with parameters (for example /item/:id) and pages that require an account.
React Context	Data shared by the whole application: the signed-in user, the cart, the favorites, the library and the admin role.
CSS	Styling and the responsive layout, using Flexbox, Grid and media queries.
Firebase Authentication	Sign-up and sign-in with email and password or with a Google account, and password reset.
Cloud Firestore	The database. It stores the books, user data, reviews and the community feed, and sends live updates to the page.
Firestore Security Rules	Rules that decide who can read or change each piece of data.
Node.js and Express	The server that exposes the API used by the AI features.
Google Gemini API	The language model used for the story maps, character chat, Plot Detective and the assistant.
firebase-admin	Used on the server to check that a request comes from a signed-in user.

Zod	Validation of the data received by the server and of the answers returned by the AI.
Firebase CLI	Publishing the security rules to the Firebase project.

3. How the Application Is Organized

The React application runs in the browser and talks directly to Firebase: it signs users in through Firebase Authentication and reads and writes data in Cloud Firestore.

The AI features cannot call Gemini directly from the browser, because the API key would be visible to anyone. Instead, the browser sends a request to the Express server, together with the user's Firebase sign-in token. The server checks the token, builds the request for Gemini and sends the answer back. For the chat features the answer is sent in pieces, so the text appears on the screen while it is being written.

The source code is organized as follows:

- src contains the React pages and components, the shared state and the functions that read and write data;
- server contains the Express server with the AI endpoints;
- firestore.rules contains the database security rules;
- scripts contains the scripts used once to fill the catalog with books.

4. Features

4.1 Home page

The home page shows the whole catalog as a grid of book cards. Each card has the cover, the main genre, the title, the author, the year and the price, plus a heart button for adding the book to favorites. If the user already owns the book, the price is replaced by "In your library".

- The search bar filters the books while the user types, by title or author.
- The books can be filtered by genre. There are nine genres, and each one shows how many books it has.
- The books can be sorted by title, by price or by year.
- At the bottom of the page there is a community feed with the latest progress updates shared by readers. It updates by itself when someone posts something new.

4.2 Navigation bar

The navigation bar stays at the top of every page. It has links to the home page, By Vibe, My List, Favorites and the cart, which shows the number of items in it. A signed-in user sees their name and picture, which lead to the account page, and a logout button. The administrator also sees a link to the admin page. On a phone the links are hidden behind a menu button.

4.3 Sign in and account creation

The sign-in page has two tabs, one for signing in and one for creating an account. Users can sign in with their Google account or with an email address and a password.

- When creating an account, the user also enters their name, and the password must have at least six characters.
- The "Forgot password?" link sends an email for resetting the password.
- Errors are shown as plain messages, for example when the email and password do not match.
- After signing in, the user is taken back to the page they were on, for example the checkout.

4.4 Book page

Each book has its own page with the cover, genres, author, year, description and average rating. From here the user can buy the book or rent it for 30 days of online reading, add it to favorites or to My List. If the user already has access to the book, a "Read now" button is shown.

- The reading progress is saved by chapter. The user chooses the last chapter they finished, and the titles of the next chapters stay hidden so that they do not give anything away.
- The user can share their progress in the community feed, with an optional comment.
- Signed-in users can leave a rating from one to five stars and a review. Each user can have one review per book, which they can update later.
- The page also links to the map, the character chat and, for crime novels, Plot Detective.

4.5 Book by Vibe

This page recommends books based on the atmosphere the user is looking for instead of the genre. The user selects options such as rain, small town, mystery, cozy or friendship, or picks one of four ready-made combinations. The books are then listed in order of how well they match, and each one shows the percentage and the options it matched.

4.6 Cart and checkout

The cart lists the chosen books, with the type of purchase (buying the book or online access) and the price, and calculates the total. Books can be removed from the cart at any time.

Checkout requires an account, because purchased books are saved on the user's account. The user enters their name and email and chooses a payment method. The payment is simulated, so no card details are requested. After the order, purchased books appear in the user's library and rented books get an expiry date 30 days later.

4.7 Online reader

Books that the user owns or rents can be opened in an online reader with a book-like layout. The reader can make the text larger or smaller and switch between a day and a night mode. Users who do not have access to the book see a message with a link to the buying options.

4.8 Favorites and My List

The Favorites page shows all books marked with the heart. My List is a separate reading list for the books the user plans to read next, and books are added to it from their page. Both lists are saved on the user's account, so they are available on any device. A guest's cart and favorites are kept in the browser and are moved to their account when they sign in.

4.9 Account page

The account page shows the user's name, picture, email and the date the account was created. From here the user can change their name, request a password change and log out.

- A short summary shows how many books the user owns, rents, has in favorites and in the cart.
- The "Continue reading" section lists the books in progress with the chapter reached.
- The library lists the purchased and rented books, each with a button for opening it in the reader.

4.10 Enter the Book

Enter the Book shows a map of the world of a book, with its places, characters and events. The map is generated by the AI the first time the book is opened and then saved in the database, so the next readers get it immediately.

The map depends on the user's progress. Places, characters and events from chapters the user has not read yet are hidden and appear as the user marks more chapters. Clicking a place shows its description, the characters met there and what happened there. Below the map there is a list of the characters met so far and a timeline of the events.

4.11 Character Chat

Character Chat lets the user talk to the characters they have already met in the book. The AI answers in the character's voice and only knows what happened up to the user's current chapter, so questions about later events are avoided in a natural way. The answer appears while it is being written, and the characters reply in the language the user writes in.

4.12 Plot Detective

Plot Detective is available for crime and mystery novels. It shows a board with the suspects and the clues the reader has found so far, which unlock chapter by chapter.

- The cards can be moved around the board and connected with a red string.
- The user can add their own notes and mark one suspect as the main suspect.
- The user writes a theory about who did it and why.
- At the end, the AI compares the theory with the real solution and gives a score from 0 to 100, with an explanation of what the user got right and what they missed.

4.13 AI assistant

A small button in the corner of every page opens a chat with the BookFlix assistant. It recommends books from the catalog, with links to their pages, and explains how the site works. It also knows what the user has in the cart and in favorites. The assistant is available to guests as well.

4.14 Admin page

The admin page can only be opened by the administrator account. Other users get a message that they do not have access.

- The overview shows the number of books, users, reviews, posts and sales, together with the most popular genres, the best-selling books and the best-rated books.
- The administrator can add new books, edit existing ones and delete them. The form checks the data before saving, for example that the title and author are filled in and that the cover is a valid link.
- The administrator can delete reviews and feed posts that are not appropriate.

5. Security

The security of the data does not depend only on hiding buttons in the interface, because anyone can see the Firebase configuration in the browser. For this reason the access rules are written in Firestore Security Rules, which are checked by Firebase for every request.

- Anyone can read the books, the reviews and the community feed.
- Only the administrator can add, change or delete books.
- Each user can read and change only their own data, such as the cart, the progress and their lists.
- Reviews and feed posts can only be written in the user's own name.
- The list of administrators can only be changed from the Firebase console.

After the rules were published, they were tested with a user who was not signed in. This user could read the catalog, but could not change a book, make themselves an administrator, see the list of users or read another user's data.

On the server side, the Gemini API key is stored only on the server, in a file that is not uploaded to GitHub. Every AI request must include a valid Firebase token, the number of requests per user is limited, and all received data is validated before it is used.

6. Running the Project

To run the project locally, Node.js and a Gemini API key from Google AI Studio are needed.

- Install the dependencies with npm install.
- Create the file server/.env, starting from server/.env.example, and write the Gemini key in it.
- Start the AI server with npm run server.
- In a second terminal, start the website with npm start. It opens at http://localhost:3000.

The source code is available at https://github.com/Cristina-hub04/BookFlix.